

Executive Business Report

Intelligent Sales Forecasting and Demand Analysis System

Executive Summary

This project was developed to help improve inventory planning and sales forecasting for a retail business. Historical sales data from four years was analyzed to identify trends, seasonal patterns, unusual sales events, and product demand behavior. Three forecasting models were evaluated, and the XGBoost model provided the most accurate predictions. The system also identified abnormal sales weeks using anomaly detection techniques and grouped products into demand-based clusters to support inventory decisions. An interactive dashboard was created to allow managers to monitor sales performance, forecasts, anomalies, and product demand in a simple and user-friendly way.

Key Findings from Data Analysis and Forecasting

Sales Performance

- Technology and Office Supplies generated the highest contribution to total revenue.
- Sales showed a clear yearly growth trend with noticeable seasonal peaks.
- Demand increased significantly during festive and holiday periods, especially in the final quarter of each year.
- The average shipping time remained consistent across regions, indicating a stable order fulfillment process.

Time Series Analysis

The decomposition of monthly sales revealed:

- A steady upward sales trend over the four-year period.
- Strong yearly seasonality, indicating recurring demand patterns.
- A few months showed unusually high residual values, suggesting unexpected sales spikes caused by external business events.

The Augmented Dickey-Fuller (ADF) test confirmed that the monthly sales series was stationary, meaning the sales pattern was stable enough for reliable forecasting without additional differencing.

Three-Month Sales Forecast

Three forecasting approaches were evaluated:

Model	MAE	RMSE	MAPE
SARIMA	19,244.49	19,950.07	20.53%
Prophet	20,250.79	22,318.41	21.86%
XGBoost	13,915.32	18,893.85	13.29%

The XGBoost model achieved the lowest forecasting error and was selected for deployment.

Forecast Summary

The model predicts that overall sales will remain stable during the next three months, with continued strength in the Technology category and the West region. The forecasts suggest that demand will remain healthy without major declines, allowing inventory planning to proceed with confidence.

Top Three Sales Anomalies

The anomaly detection models identified several unusual sales weeks. The three most significant anomalies include:

1. March 22, 2015

A sharp increase in weekly sales was detected by both Isolation Forest and Z-Score methods. This may indicate a major promotional campaign or special sales event.

2. November 2018

Multiple unusually high sales weeks occurred during November, which likely correspond to festive season shopping and promotional offers.

3. December 2017

A significant sales spike was observed during the holiday season, suggesting increased customer demand before year-end.

These anomalies represent important business events that should be considered when planning future inventory and marketing campaigns.

Product Demand Segmentation

Product sub-categories were grouped using K-Means clustering based on:

- Total Sales
- Sales Growth Rate
- Sales Volatility
- Average Order Value

Four meaningful demand groups were identified.

Demand Segment	Recommended Strategy
High Volume, Stable Demand	Maintain high inventory levels to prevent stock shortages.
Growing Demand	Gradually increase inventory as demand continues to rise.
Low Volume, High Volatility	Keep moderate inventory and monitor sales frequently.
Declining Demand	Reduce inventory and avoid overstocking.

The West region demonstrated the strongest expected sales growth over the next three months and should receive additional inventory allocation.

Business Recommendations

1. Increase Inventory in High-Growth Segments

The forecast indicates strong future demand in the West region and Technology category. Increasing inventory for these segments can reduce stock-outs and improve customer satisfaction.

2. Prepare Inventory Before Seasonal Peaks

Historical analysis shows that sales consistently rise during festive and holiday periods. Increasing stock levels before these periods can improve product availability and maximize revenue.

3. Monitor Sales Anomalies Regularly

Unexpected spikes and drops identified through anomaly detection should be reviewed promptly. Early identification allows management to investigate promotions, supply chain issues, or unexpected customer demand before they impact operations.

Risk and Limitation

The forecasting models rely entirely on historical sales data. Unexpected events such as economic changes, supply chain disruptions, competitor actions, or new product launches are not included in the predictions. As a result, forecasts should be updated regularly using the latest sales data to maintain accuracy.

Conclusion

This project successfully demonstrates an end-to-end sales forecasting and inventory planning solution. By combining time series forecasting, machine learning, anomaly detection, product segmentation, and an interactive dashboard, the system provides practical insights that can support better inventory management, improve demand forecasting accuracy, and enable more informed business decisions. The XGBoost forecasting model achieved the highest prediction accuracy and is recommended for production use due to its superior performance on this dataset.